The Fundamental Theorem of Calculus

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Calculus II · STM 1002 · University of Austin

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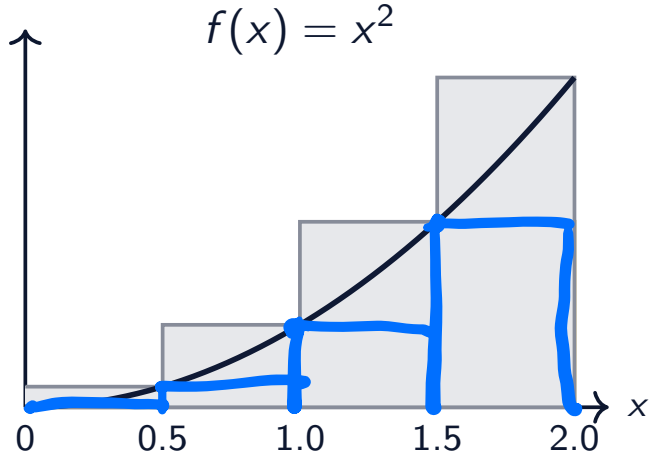
Thursday's Riemann Sums

$f(x) = x^2$ on $[0, 2]$, $n = 4$

- $L_4 = 1.75$
- $R_4 = 3.75$

The exact answer is somewhere in between...



$L_4 = 1.75$
$R_4 = 3.75$

$n \longrightarrow \infty$

we get to the precise answer

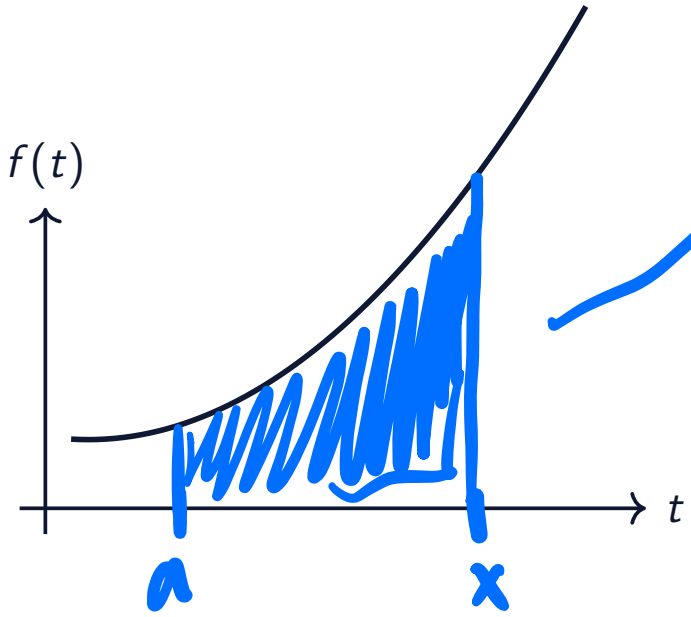
The Accumulation Function

$$F(x) = \int_a^x f(t)\,dt$$

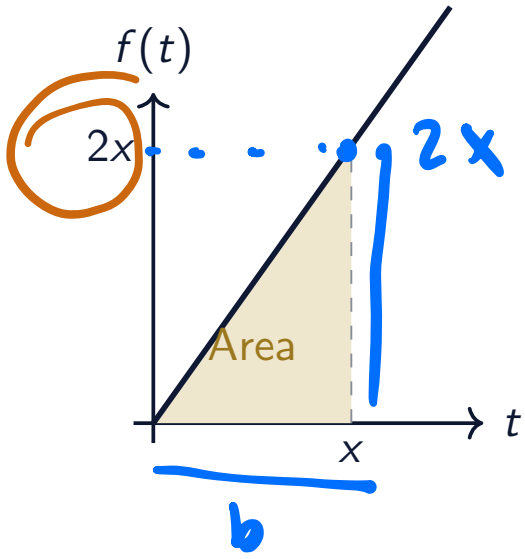
$\int_a^x f(t)\,dt$

$F(x)$

A function of x: the **running total** of area under f from a to x.



Example: $f(t) = 2t$



$$F(x) = \int_0^x 2t\,dt \quad = x^2$$

Find $F(x)$ using geometry. Then compute $F'(x)$.

Area of a triangle $= \frac{1}{2} \cdot b \cdot h$

$$F(x) = \int_0^x 2t\,dt$$

$$= \frac{1}{2} \cdot x \cdot 2x = \frac{2x^2}{2} = \boxed{x^2} = F(x)$$

$$F'(x) = 2x$$

The derivative of the accumulated area

equals the **original function**.

$$F'(x) = f(x)$$

when is this true?

FTC Part 1

The Fundamental Theorem of Calculus, Part 1

If f is continuous on $[a, b]$, then the function

$$F(x) = \int_a^x f(t)\,dt$$

satisfies $\quad F'(x) = f(x) \quad$ for all $x \in (a, b)$.

Differentiation undoes integration.

The accumulation function F is an **antiderivative** of f.

Antiderivatives

power rule derivative

$x^n = nx^{n-1}$

An **antiderivative** of f is any function F with $F' = f$.

$x^n = \frac{x^{n+1}}{n+1}$

$f(x)$		An antiderivative $F(x)$
$2x$	$\longrightarrow$	$x^2 + C$
x^2	$\longrightarrow$	$\frac{x^3}{3}$
e^x	$\longrightarrow$	e^x

$x^2 + 3$

$x^2 + 7$

FTC Part 2

The Fundamental Theorem of Calculus, Part 2

If $F' = f$ on $[a, b]$, then

$$\int_a^b f(x)\,dx \;=\; F(b) - F(a)$$

Bracket notation:

$$\Big[F(x)\Big]_a^b \;=\; F(b) - F(a)$$

Find an antiderivative, plug in endpoints, subtract.

From Last Time: $\int_0^2 x^2\,dx$

$x^3 = 2^3$

$$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3} - 0 = \frac{8}{3} \approx 2.667$$

Method	Value	
Left sum (Wed)	1.75	underestimate
Right sum (Wed)	3.75	overestimate
FTC (today)	2.667	in between

Example 2: $\int_0^1 e^x \, dx$

$$\int_0^1 e^x dx = \left[e^x\right]_0^1 = e^1 - e^0 = \underline{e - 1}$$

$$= 1.718$$

Compare: L_4 from Thursday ≈ 1.5125

Practice 1

Evaluate using FTC Part 2:

$$\int_1^3 2x\,dx$$

$$\left[x^2\right]_1^3 = 9 - 1 = 8$$

Practice 2

$$\int \sin(x) = -\cos(x)$$

Evaluate using FTC Part 2:

$$\int_0^{\pi} \sin(x)\, dx$$

$$\left[-\cos(x)\right]_0^{\pi} = \underbrace{-\cos(\pi)} - \left(\underbrace{-\cos(0)}\right)$$

$$-(-1) - (-1) = 2$$

Practice 3

Evaluate using FTC Part 2:

$$\int_1^4 (3x^2 + 2x)\, dx$$

$$\int_1^4 3x^2\, dx + \int_1^4 2x\, dx$$

$$x^3 + x^2$$

Practice 4

Evaluate using FTC Part 2. Can you explain the result?

$$\int_{-1}^{1} x^3 \, dx$$

$$\left[\frac{x^4}{4}\right]_{-1}^{1} = \frac{1}{4} - \frac{1}{4} = 0$$

Practice 5: Accumulation in Context

Water flows into a tank at rate $r(t) = 6t$ liters/min.

How much water enters between $t = 0$ and $t = 3$?

Antiderivatives So Far

$f(x)$	$F(x)$
$x^n \quad (n \neq -1)$	$\dfrac{x^{n+1}}{n+1}$
e^x	e^x
$\sin x$	$-\cos x$
$\cos x$	$\sin x$

Announcements

- Homework 1 is posted
- Practice problems are posted (with solutions). Not required.
- Office Hours are W 9:30 AM - 11:30 AM, Second Floor, otherwise by appointment.
  - Email me for second floor access as needed.
- Coming Monday: more antiderivatives
- Shoot me an email if you ever have any issues with the course, or questions about the material!